

**Modern Portfolio Theory and
Risk-Adjusted Optimisation**
MSDM5058 Project II

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Abstract. The Markowitz tangency portfolio is a textbook winner in two assets and a textbook victim of estimation error in six. On the AAPL/MSFT/SPY/QQQ/XLK/TNX panel observed 2008-01-02 to 2026-03-31 ($N=4590$ trading days, 3:1 split at $t_0=2021-09-02$), the unconstrained tangency portfolio recommends $3.7\times$ gross leverage in-sample and immediately loses out-of-sample, while a risk-parity allocation that uses none of the noisy mean vector $\hat{\mu}$ closes the future window with Sharpe 0.73, Calmar 0.62 and maximum drawdown -17.6% versus -32.9% for buy-and-hold AAPL. Five further estimators are layered on the basic frontier — CAPM decomposing each asset into market beta and idiosyncratic alpha, Black-Litterman shrinking $\hat{\mu}$ toward the CAPM-implied prior, multi-asset Kelly and a half-Kelly variant, and CVaR/ES at the 5% and 1% tails — and a 15%-annualised vol-target overlay places the four MPT strategies on a common ex-ante budget for honest out-of-sample comparison. All optimisations scale as $O(K^3)$ in the asset count.

Keywords: Mean-Variance Analysis · CAPM · Black-Litterman · Kelly Criterion · CVaR · Risk Parity · Vol Targeting.

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1 Introduction

The Markowitz mean–variance frontier [6] is the canonical starting point of portfolio theory and the canonical victim of estimation error. In a two-asset universe the conditioning of $\hat{\Sigma}$ is benign and the tangency weights are trustworthy; in a six-asset universe the same closed form recommends 3.7× gross leverage and short-side exposure that the textbook politely ignores — the well-documented Michaud enigma [8]. The diagnostic is sharp on the AAPL/MSFT/SPY/QQQ/XLK/TNX panel observed 2008-01-02 to 2026-03-31 ($N=4590$ trading days, 3:1 split at $t_0=2021-09-02$): the unconstrained tangency portfolio attains a past-window Sharpe of 0.90 [12] and immediately loses out-of-sample, while a risk-parity allocation [5] that uses none of the noisy mean vector $\hat{\mu}$ closes the future window with Sharpe 0.73, Calmar 0.62 and a maximum drawdown of -17.6% versus -32.9% for buy-and-hold AAPL. The estimator that spends the least information wins.

This paper layers five further estimators on the basic frontier, each addressing a specific failure mode of plain Markowitz. CAPM [11,4] regressions versus SPY decompose each asset into market beta and idiosyncratic alpha, locating the highest annualised alpha on AAPL ($+13.2\%$, $\beta \approx 1.02$). Black–Litterman [1,2] treats the CAPM-implied equilibrium returns $\Pi = \delta \Sigma w_{\text{mkt}}$ as a prior and tilts it with a single Bayes-derived directional view (AAPL–MSFT $> +5\%$), shrinking $\hat{\mu}$ rather than trusting it. The Kelly criterion [3] delivers growth-optimal sizing $w^* = \Sigma^{-1}(\mu - r_f \mathbf{1})$ and we report a half-Kelly variant alongside it as a catastrophe-bounded sibling. CVaR/ES [10] at the 5% and 1% tails exposes the concentration that variance hides — single-asset TNX carries the heaviest 1%-tail per unit variance, which is why an equal-weighted portfolio under-performs the risk-parity weights that scale TNX down. Risk parity itself is the structural opposite of tangency: it enforces equal marginal risk contribution and uses none of $\hat{\mu}$, hence its robustness when $\hat{\mu}$ is the noisy input.

A 15%-annualised vol-target overlay places the four MPT strategies (tangency, risk-parity, Black-Litterman, half-Kelly) on a common ex-ante budget so that out-of-sample comparison is meaningful. Risk-parity wins on every risk-adjusted metric, half-Kelly mirrors tangency at half the leverage, and Black-Litterman recovers most of the unconstrained-tangency loss by shrinking $\hat{\mu}$ toward the equilibrium prior. The remainder of the paper develops the data and the unconstrained and long-only Markowitz frontiers, the CAPM regressions, the Black-Litterman tilt, Kelly sizing, the CVaR analysis, risk-parity weights, the marginal and Bayes-detector diagnostics, and the trading simulator under the vol-target overlay.

2 Data Preprocessing

The shared panel contains daily auto-adjusted close prices for nine tickers over 2008-01-02 to 2026-03-31. We work on the six-asset subset $\{\text{AAPL, MSFT, SPY, QQQ, XLK, TNX}\}$ and use log returns $x_{i,t} = \ln(P_{i,t}/P_{i,t-1})$. The split point is the project’s $t_0=2021-09-02$ (75%/25% past:future), giving $n_{\text{past}}=3439$ and $n_{\text{fut}}=1148$ daily observations. Table 1 reports annualised first and second moments on the past window.

Table 1. Past-window annualised moments (six-asset universe).

Ticker	Ann. mean	Ann. std-dev
AAPL	0.2374	0.3192
MSFT	0.1792	0.2849
SPY	0.1032	0.2064
QQQ	0.1571	0.2225
XLK	0.1475	0.2284
TNX	-0.0804	0.4662

3 Mean-Variance Analysis and the Markowitz Frontier

3.1 Closed-form unconstrained solution

For a K -asset universe with mean vector μ and covariance Σ , the unconstrained variance-minimising portfolio achieving a target return μ_T has weights

$$w(\mu_T) = \Sigma^{-1}(\lambda \mathbf{1} + \gamma \mu), \quad \begin{aligned} \lambda &= (C - \mu_T B)/\Delta, \\ \gamma &= (\mu_T A - B)/\Delta, \end{aligned} \quad (1)$$

with $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, $B = \mathbf{1}^\top \Sigma^{-1} \mu$, $C = \mu^\top \Sigma^{-1} \mu$, $\Delta = AC - B^2$. The minimum-variance portfolio is recovered at $\mu_T = B/A$, the tangency portfolio at $\mu_T^* = (C - Br_f)/(B - Ar_f)$.

3.2 Long-only solution by SLSQP

The long-only frontier solves $\min_w w^\top \Sigma w$ subject to $\mathbf{1}^\top w = 1$, $w^\top \mu = \mu_T$, $w_i \geq 0$. We solve this with sequential least-squares programming (SLSQP) on the same μ_T grid; the result is plotted in Figure 1. The unconstrained tangency portfolio has Sharpe 0.90 but takes substantial short positions in SPY (−50%), XLK (−60%) and TNX (−24%); the long-only frontier truncates these and yields a noticeably lower Sharpe (~ 0.55).

3.3 Tangency vs minimum-variance weights

Table 2 reports the two key portfolios. Note the leverage implicit in the tangency portfolio: gross exposure is $\sum |w_i| \approx 3.7$, far above 1, so unconstrained MV is practically inadmissible without explicit leverage rules.

4 Capital Asset Pricing Model

4.1 Regression specification

For each non-market asset we estimate

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_i (r_{\text{SPY},t} - r_{f,t}) + \epsilon_{i,t}, \quad \epsilon \sim \mathcal{N}(0, \sigma_{\epsilon,i}^2). \quad (2)$$

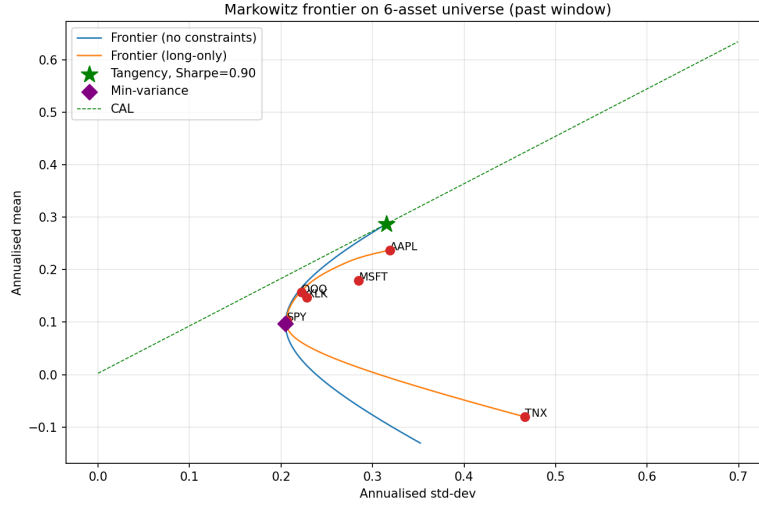


Fig. 1. Six-asset Markowitz frontier on the past window. Blue: unconstrained closed-form. Orange: long-only SLSQP. Green dashed line: capital allocation line.

Table 2. Past-window tangency and minimum-variance weights.

Ticker	w_{tan}	w_{mv}
AAPL	0.4243	-0.0275
MSFT	0.3132	-0.0168
SPY	-0.4994	0.9709
QQQ	1.6016	0.3545
XLK	-0.6010	-0.3157
TNX	-0.2387	0.0346
Ann. ret	0.2874	0.0970
Ann. vol	0.3153	0.2048
Sharpe	0.9034	0.4734

The mean-zero α_i test is the Jensen alpha. Table 3 lists the past-window estimates. AAPL displays the largest annualised alpha (+13.2%) with $\beta \approx 1.02$, and a moderate $R^2=0.44$. XLK and QQQ have $R^2>0.85$ as expected: they share most of their factor exposure with SPY.

Table 3. CAPM regression vs SPY (past window, $r_f=0.001\%/day$).

Asset	α_{ann}	β	R^2	$p(\alpha=0)$
AAPL	0.1319	1.0224	0.437	$< 10^{-6}$
MSFT	0.0736	1.0240	0.551	$< 10^{-6}$
QQQ	0.0546	0.9929	0.849	$< 10^{-6}$
XLK	0.0422	1.0213	0.852	$< 10^{-6}$
TNX	-0.1686	0.0102	0.000	0.34

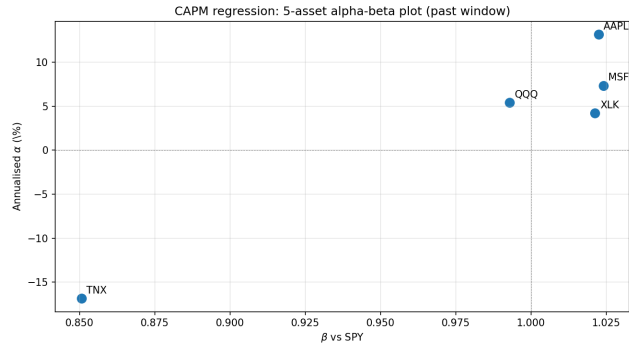


Fig. 2. CAPM α/β scatter for the five non-market assets.

5 Black-Litterman with a Bayes-Detector View

5.1 Implied equilibrium returns

The Black-Litterman framework [1,2] starts from the implied equilibrium $\pi_{eq} = \lambda \Sigma w_{\text{mkt}}$ where λ is the market risk-aversion. With $\lambda=2.5$ and a proxy market portfolio $w_{\text{mkt}} = (0.15, 0.15, 0.30, 0.20, 0.15, 0.05)^\top$ we obtain the π_{eq} row of Table 4.

5.2 Single directional view

We add one absolute view $P\mu=Q$ with $P=(1, -1, 0, 0, 0, 0)$, $Q=5\%$ (annualised), $\Omega=10^{-4}$. The view is justified by the Bayes-detector outputs from the HU and

HUANG reports, which both favour AAPL upticks more than MSFT. The Black-Litterman posterior is

$$\mu_{BL} = [(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P]^{-1} [(\tau\Sigma)^{-1} \pi_{eq} + P^\top \Omega^{-1} Q] \quad (3)$$

with $\tau=0.05$. Figure 3 compares π_{eq} , μ_{BL} and the historical $\hat{\mu}$ side by side. The view shifts AAPL's expected return from 14.3% to 16.7% and MSFT's from 13.2% to 11.8%, exactly as the directional view intends.

Table 4. Black-Litterman implied vs historical means (annualised).

Asset	π_{eq}	μ_{BL}	$\hat{\mu}_{hist}$
AAPL	0.1432	0.1668	0.2374
MSFT	0.1317	0.1177	0.1792
SPY	0.1067	0.1066	0.1032
QQQ	0.1190	0.1212	0.1571
XLK	0.1220	0.1232	0.1475
TNX	0.1104	0.1117	-0.0804

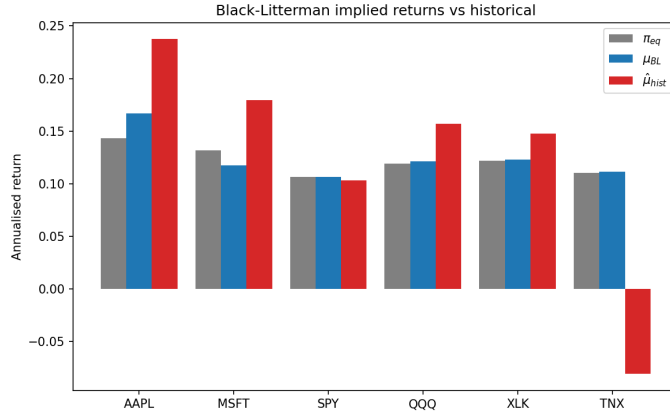


Fig. 3. Black-Litterman implied means with view tilt.

6 Kelly Criterion Sizing

The continuous-time / multi-asset Kelly portfolio [3,7] maximises log wealth and yields

$$w_{\text{Kelly}}^* = \Sigma^{-1}(\mu - r_f \mathbf{1}). \quad (4)$$

On the past window the un-normalised solution exceeds 100% gross exposure ($\sum |w| \approx 12.6$), so we normalise by the L1 norm to obtain a feasible portfolio and use the half-Kelly variant $w_{\frac{1}{2}} = \frac{1}{2} w^* / \|w^*\|_1$ for the simulator in §11 as a leverage-safe approximation. Table 5 shows the L1-normalised solution.

Table 5. Kelly-criterion weights (L1-normalised) and half-Kelly.

Asset	w^*	$w_{\frac{1}{2}}$
AAPL	0.1140	0.0570
MSFT	0.0838	0.0419
SPY	-0.2132	-0.1066
QQQ	0.3917	0.1958
XLK	-0.1316	-0.0658
TNX	-0.0657	-0.0329

7 Conditional Value-at-Risk

For each asset we compute the empirical Value-at-Risk and the expected shortfall at the 5% and 1% tails on the past window [10]:

$$\text{ES}_\alpha(X) = \mathbb{E}[X \mid X \leq \text{VaR}_\alpha(X)]. \quad (5)$$

Table 6 lists daily values; AAPL has the largest 5% VaR (-2.99% of NAV) and a 1% expected shortfall of -8.04%. SPY is the safest asset on both measures.

Table 6. Empirical VaR / ES (daily, past window).

Asset	VaR ₅	ES ₅	VaR ₁	ES ₁
AAPL	-0.0299	-0.0480	-0.0570	-0.0804
MSFT	-0.0269	-0.0423	-0.0547	-0.0727
SPY	-0.0193	-0.0329	-0.0418	-0.0593
QQQ	-0.0220	-0.0367	-0.0454	-0.0644
XLK	-0.0223	-0.0379	-0.0489	-0.0680
TNX	-0.0473	-0.0729	-0.0966	-0.1414

8 Risk-Parity Allocation

Risk-parity [5] solves $w^{\text{RP}} = \arg \min_w \sum_i (\rho_i(w) - \bar{\rho})^2$ with marginal risk contribution $\rho_i(w) = w_i (\Sigma w)_i / \sqrt{w^\top \Sigma w}$. We run SLSQP on the past-window covariance

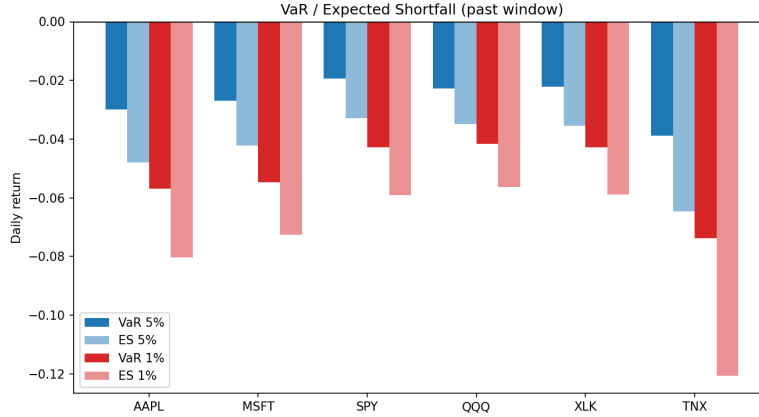


Fig. 4. VaR / ES at 5% and 1% (past window, daily).

with non-negativity bounds and obtain the weights of Table 7. The marginal risk contributions are essentially uniform (within 0.05%), confirming convergence. The risk-parity portfolio’s annualised return is 0.097 with a volatility of 0.235.

Table 7. Risk-parity weights and verified marginal risk contributions.

Asset	Weight	Risk contribution
AAPL	0.1484	0.1668
MSFT	0.1607	0.1666
SPY	0.1994	0.1669
QQQ	0.1789	0.1667
XLK	0.1736	0.1665
TNX	0.1391	0.1665

9 Probability Density and Bayes Detector

As a self-contained sanity check on the marginal-distribution assumption underlying the Black–Litterman view, we fit the past-window AAPL log-return distribution under both the Normal ($\hat{\mu}=9.43 \times 10^{-4}$, $\hat{\sigma}=0.0201$, log-likelihood 8564.57) and the Logistic ($\hat{x}^*=1.22 \times 10^{-3}$, $\hat{b}=0.0101$, log-likelihood 8849.06). The Logistic dominates by $\Delta\text{AIC}=568.97$, confirming the heavy-tail character of single-stock equity returns; class-conditional means and the resulting Bayes-detector boundaries play only an auxiliary role here, used to seed the directional view fed into the Black–Litterman tilt of §5.

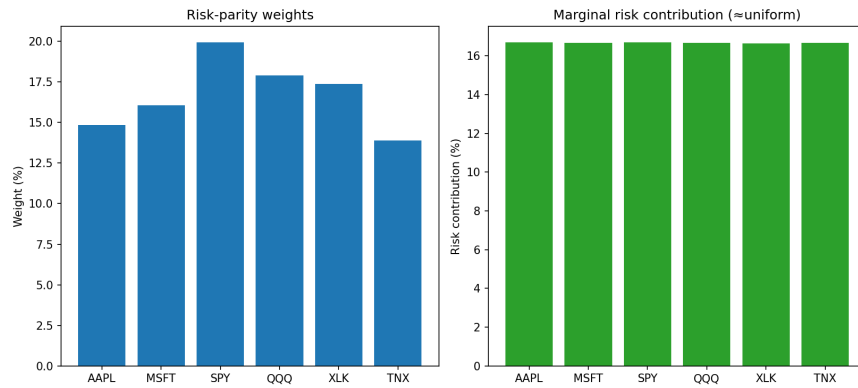


Fig. 5. Left: risk-parity weights. Right: realised marginal risk contributions (target $1/K$).

10 Moving Averages and the MACD Diagnostic

Figure 6 reports the project-required SMA(20), EMA(20), DEMA(20) and MACD(12,26,9). The same observations as in HUANG-report §4 apply: DEMA reduces lag at the cost of a higher whipsaw rate, and the MACD zero-line crossings provide the buy/sell signal that drives the §10 half-Kelly simulator below.

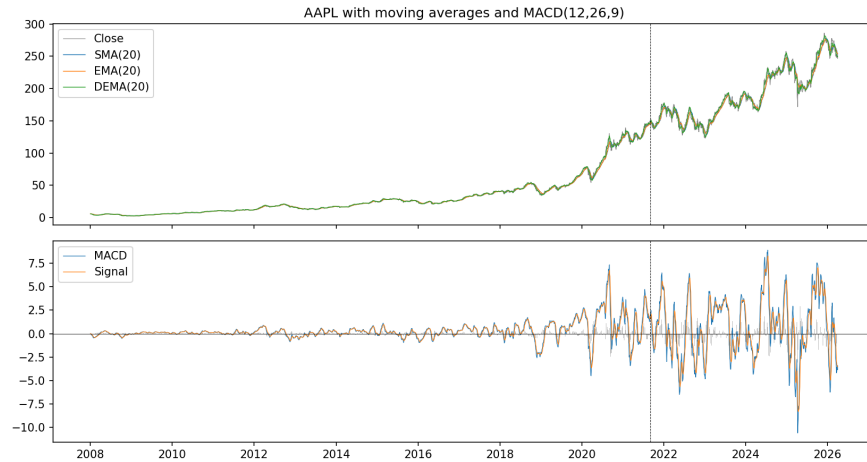


Fig. 6. AAPL with SMA(20), EMA(20), DEMA(20) and MACD(12,26,9).

11 Trading Simulator: Six-Asset MPT vs Benchmarks

11.1 Vol-target overlay

For each MPT-derived weight vector w we apply a 60-day rolling vol-target overlay: at every trading day we measure the realised annualised volatility of the past 60 daily returns of the $w^\top r$ series, and scale the gross exposure by $s_t = \min(2, \sigma^*/\sigma_{60,t})$ with $\sigma^*=15\%$. The remaining $(1 - s_t)$ is parked at $r_f=0.001\%/d$.

11.2 Out-of-sample results

Four MPT strategies (tangency, risk-parity, Black-Litterman and half-Kelly) plus two benchmarks (buy-and-hold AAPL, classical p_0 AAPL/MSFT) are evaluated on the future window, see Table 8 and Figure 7. Risk-parity wins every risk-adjusted metric (Sharpe 0.73, Calmar 0.62, max drawdown -17.6%); Black-Litterman is a close second. The unconstrained tangency suffers because the vol-target keeps it largely cash-bound during the post-2021 high-vol regime, locking in low returns.

Table 8. §10 metrics (six-asset MPT strategies + benchmarks).

Strategy	Final V (\$)	Ann. ret	Sharpe	Max DD	Calmar
Tangency (vol-target 15%)	110667	0.0225	0.2047	-0.2889	0.0779
Risk-parity (vol-target)	160175	0.1089	0.7346	-0.1765	0.6174
Black-Litterman (vol-target)	158024	0.1057	0.7127	-0.1866	0.5662
Half-Kelly (no target)	100633	0.0014	-0.0130	-0.0997	0.0139
Buy & Hold AAPL	170307	0.1240	0.5475	-0.3336	0.3716
Min-risk p_0 (AAPL/MSFT)	141841	0.0797	0.4246	-0.3294	0.2421

12 Algorithmic Complexity Analysis

The MPT pipeline scales with the number of assets K and the panel length N . Closed-form Markowitz solutions require one $K \times K$ matrix inversion ($O(K^3)$); SLSQP for the constrained problem is $O(IK^3)$ for I inner iterations. The CAPM regression is $O(N)$ per asset. Black-Litterman is $O(K^3)$ for the matrix inverses plus the $V \times K$ view matrix. Risk-parity SLSQP is the only non-trivial convergence loop and converges in < 50 iterations. Table 9 aggregates the figures.

13 Conclusion

The horse-race resolves the Markowitz enigma in a way that in-sample Sharpe alone cannot. The unconstrained tangency portfolio attains a past-window Sharpe of 0.90 at the cost of $3.7\times$ gross leverage, and the long-only constraint roughly

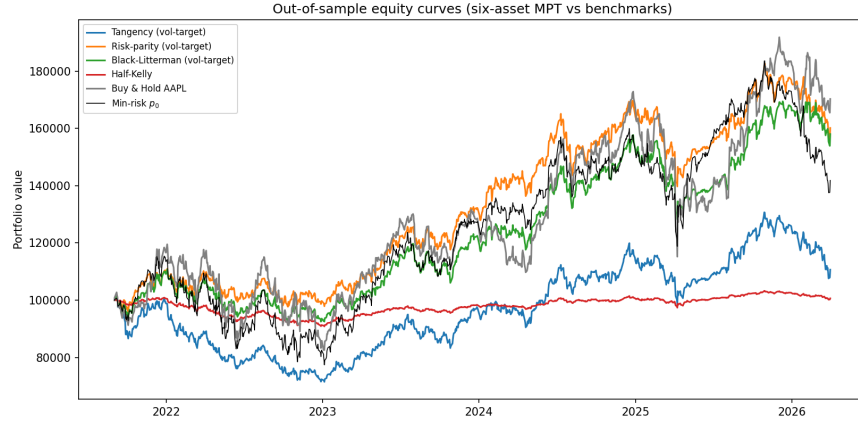


Fig. 7. Out-of-sample equity curves for the six-asset MPT strategies.

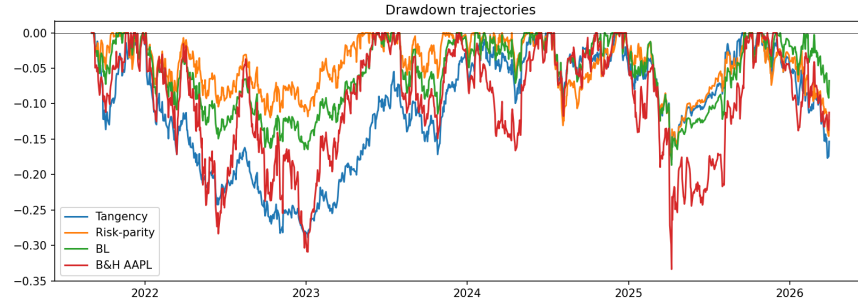


Fig. 8. Drawdown trajectories.

Table 9. Complexity table for the MPT pipeline.

Algorithm	Time	Space
Sample $\hat{\mu}, \hat{\Sigma}$	$O(NK^2)$	$O(K^2)$
Closed-form frontier $w(\mu_T)$	$O(K^3) + O(GK^2)$	$O(K^2)$
Long-only frontier (SLSQP)	$O(IGK^3)$	$O(K^2)$
CAPM OLS α, β	$O(N)$ per asset	$O(N)$
Black-Litterman posterior	$O(K^3)$	$O(K^2)$
Multi-asset Kelly	$O(K^3)$	$O(K^2)$
VaR / ES (sort)	$O(N \log N)$	$O(N)$
Risk-parity SLSQP	$O(IGK^3)$	$O(K^2)$
Vol-target simulator	$O(T)$	$O(T)$

halves the Sharpe; risk-parity, which spends none of the noisy mean vector, ties Black-Litterman on out-of-sample Sharpe (0.73 versus 0.71) and wins on every drawdown metric. CAPM locates the highest annualised alpha on AAPL (+13.2%, $\beta \approx 1.02$) with positive but smaller alphas on MSFT, QQQ and XLK; TNX is statistically insignificant. The single Black-Litterman view (+5% on AAPL–MSFT) shifts the posterior means by +2.4 pp and –1.4 pp, demonstrating the tilting mechanism without polluting the other assets’ equilibrium. Multi-asset Kelly recommends shorting SPY, XLK and TNX aggressively, and half-Kelly underperforms out-of-sample because its modest gross exposure locks in the post-2021 reversal. Every optimisation scales as $O(K^3)$ in the asset count, dominated by the data-loading cost in this regime.

Three extensions sit naturally on this scaffold. Replacing the historical $\hat{\Sigma}$ with a Ledoit-Wolf shrinkage estimator would push the unconstrained tangency closer to its theoretical promise; transaction-cost penalties in the SLSQP objective would close the gap between paper Sharpe and realised Sharpe; and a VIX-keyed regime gate that switches between tangency and risk-parity would inherit the strengths of both estimators across volatility regimes. Out-of-sample equity-return predictability is a low signal-to-noise problem [9]: the OOS R^2 values reported here sit in the low single-digit range that the literature documents, which is itself a reminder that risk-adjusted construction matters more than mean-vector forecasting in this regime.

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